Weak Uniqueness for Some Discontinuous Multidimensional Diffusions: A Corrected Fokker–Planck Criterion and Two Structural Partial Results

Prepared as a verification and repair note

24 June 2026

Abstract

This note repairs a proposed solution to the weak-uniqueness question raised around multidimensional Itô diffusions with discontinuous uniformly elliptic diffusion coefficient. The repaired result is *not* a solution of the full open problem. The general necessary-and-sufficient statement "weak uniqueness is equivalent to uniqueness of the associated Fokker–Planck equation" is a correct and useful standard reformulation, but by itself it is not a coefficient-level characterization and does not settle the diagonal rank/threshold diffusions arising in large ranking games.

The genuine partial progress proved here is the following. First, the martingale-problem/Fokker–Planck equivalence is stated with the missing conditioning and superposition hypotheses and proved without invoking an unjustified measurable transition kernel. Second, two checkable discontinuous classes are shown to be weakly unique in every dimension: isotropic Borel covariance matrices $A(x) = a(x)I$ with $0 < \lambda \leq a \leq \Lambda$, and coordinate-separable diagonal covariance matrices

$$A(x) = \mathrm{diag}(q_1(x_1), \ldots, q_d(x_d)), \qquad 0 < \lambda \leq q_i \leq \Lambda,$$

with arbitrary bounded Borel drift allowed in both cases. These results cover many piecewise-constant discontinuities, but not the full rank-dependent diagonal coefficients of the motivating game.

1 Status and relation to the open problem

The problem motivating this note asks for conditions, simultaneously necessary and sufficient, guaranteeing uniqueness in law of weak solutions of multidimensional Itô SDEs, especially in dimensions larger than two and for uniformly elliptic discontinuous diagonal diffusion matrices. In the ranking-game application one encounters driftless systems of the form

$$\mathrm{d}X_t = \sigma(X_t)\ \mathrm{d}W_t, \qquad A(x) = \sigma(x)\sigma(x)^\top, \tag{1}$$

where A is diagonal, uniformly elliptic, and may jump across ranking or threshold interfaces.

The repaired conclusions are as follows.

Claim	Status in this note
General necessary-and-sufficient coefficient criterion for all Borel uniformly elliptic multidimensional diffusions	Not obtained. The full problem remains open here.
Weak uniqueness iff uniqueness of the Fokker–Planck Cauchy problem	Correct under standard superposition and conditioning hypotheses, but only a reformulation.
Weak uniqueness for $A(x) = a(x)I$, a bounded Borel and uniformly elliptic	Proved, with bounded Borel drift allowed.
Weak uniqueness for $A(x) = \operatorname{diag}(q_1(x_1), \dots, q_d(x_d))$, q_i bounded Borel and uniformly elliptic	Proved, with bounded Borel drift allowed.
Rank-dependent diagonal coefficients depending on the ordering of all coordinates	Not settled by the present argument.

Thus this note should be read as partial progress plus a correction of the earlier overstatement. The equivalence in section 2 is useful because it identifies exactly what a PDE approach must prove, but the hard part is precisely to verify that Fokker–Planck uniqueness holds from assumptions on the coefficients. The time-change arguments in section 3 give such verification for two special discontinuous coefficient classes.

2 Martingale problems and Fokker–Planck equations

Fix $0 \le s < T < \infty$ and set $\Omega_s = C([s,T];\mathbb{R}^d)$ with coordinate process $X_t(\omega) = \omega(t)$ and canonical filtration $\mathcal{F}_t^s = \sigma(X_r : s \le r \le t)$. Let

$$b : [s,T] \times \mathbb{R}^d \to \mathbb{R}^d, \qquad A : [s,T] \times \mathbb{R}^d \to \mathbb{S}_+^d$$

be Borel coefficients, locally bounded for simplicity in this section, and define

$$L_t f(x) = b(t,x) \cdot \nabla f(x) + \frac{1}{2} \operatorname{Tr}\left(A(t,x) D^2 f(x)\right), \qquad f \in C_c^\infty(\mathbb{R}^d). \tag{2}$$

Definition 2.1 (Martingale problem)**.** *A probability law P on Ω_s solves the martingale problem for $(L_t)_{t\in[s,T]}$ with initial law $\nu \in \mathcal{P}(\mathbb{R}^d)$ if $P \circ X_s^{-1} = \nu$ and, for every $f \in C_c^\infty(\mathbb{R}^d)$,*

$$M_t^f := f(X_t) - f(X_s) - \int_s^t L_r f(X_r)\, \mathrm{d}r, \qquad s \le t \le T, \tag{3}$$

is a P-martingale with respect to $(\mathcal{F}_t^s)$.

Definition 2.2 (Fokker–Planck solution)**.** *A narrowly continuous curve $(\mu_t)_{t\in[s,T]} \subset \mathcal{P}(\mathbb{R}^d)$ solves the Fokker–Planck Cauchy problem from (s,ν) if $\mu_s = \nu$ and, for all $f \in C_c^\infty(\mathbb{R}^d)$ and $t \in [s,T]$,*

$$\int f \,\mathrm{d}\mu_t = \int f \,\mathrm{d}\nu + \int_s^t \int L_r f(x)\, \mu_r(\mathrm{d}x)\, \mathrm{d}r. \tag{4}$$

Equivalently, in the sense of distributions,

$$\partial_t \mu_t = L_t^* \mu_t, \qquad \mu_s = \nu.$$

Every martingale-problem solution has Fokker–Planck one-time marginals. The converse statement needed below is the superposition principle: a Fokker–Planck solution is represented by the one-time marginals of at least one martingale-problem solution. This is the Ambrosio–Figalli–Trevisan principle in the diffusion setting; bounded coefficients are more than enough for the version needed here. We use it as a hypothesis to make the logical content transparent.

Theorem 2.3 (Corrected weak-uniqueness/FPK equivalence)**.** *Assume on every subinterval $[u,T] \subset [s,T]$ that the following two standard facts hold.*

(H1) Every Fokker–Planck solution is superposed by a martingale-problem law with the same one-time marginals.

(H2) If P solves the martingale problem on $[s,T]$ and $u \in [s,T]$, then a regular conditional law of the shifted future path $(X_r)_{r\in[u,T]}$ given $\mathcal{F}_u^s$ solves the martingale problem on $[u,T]$ started from X_u, for P-a.e. conditioning path.

Then the following are equivalent.

(a) For every $u \in [s,T]$ and every $\nu \in \mathcal{P}(\mathbb{R}^d)$, the martingale problem on $[u,T]$ has at most one solution.

(b) For every $u \in [s,T]$ and every $\nu \in \mathcal{P}(\mathbb{R}^d)$, the Fokker–Planck Cauchy problem on $[u,T]$ has at most one narrowly continuous probability solution.

Consequently, whenever the martingale problem is equivalent to the weak formulation of the Itô SDE, weak uniqueness for all initial laws is equivalent to Fokker–Planck determinacy for all initial laws.

Proof. Assume first martingale-problem uniqueness. Let (μ_t) and $(\widetilde{\mu}_t)$ be two Fokker–Planck solutions from the same initial pair (u,ν). By (H1), choose martingale-problem laws P and $\widetilde{P}$ whose one-time marginals are (μ_t) and $(\widetilde{\mu}_t)$. Martingale-problem uniqueness gives $P = \widetilde{P}$, hence $\mu_t = \widetilde{\mu}_t$ for all t.

Conversely assume Fokker–Planck uniqueness for every initial pair. Let P and Q be martingale-problem solutions on $[u,T]$ with common initial law ν. Their one-time marginal curves solve (4); hence

$$P \circ X_t^{-1} = Q \circ X_t^{-1}, \qquad u \le t \le T. \tag{5}$$

We prove equality of all finite-dimensional distributions. Fix

$$u \le t_0 < t_1 < \cdots < t_n \le T$$

and bounded Borel functions $\varphi_0, \dots, \varphi_n$. The case $n = 0$ is exactly (5). Assume equality is known up to time t_{n-1}.

Let m denote the common law of $X_{t_{n-1}}$ under P and Q. Take regular conditional future laws P_x and Q_x given $X_{t_{n-1}} = x$. By (H2), for m-a.e. x these are martingale-problem solutions on $[t_{n-1},T]$ started from x. Therefore the one-time marginal curves of P_x and Q_x solve the Fokker–Planck equation from (t_{n-1},δ_x). Fokker–Planck uniqueness gives

$$g_P(x) := \mathbb{E}_{P_x}[\varphi_n(X_{t_n})] = \mathbb{E}_{Q_x}[\varphi_n(X_{t_n})] =: g_Q(x)$$

for m-a.e. x. Consequently, with $H = \prod_{k=0}^{n-1} \varphi_k(X_{t_k})$,

$$\begin{aligned}
\mathbb{E}_P\Big[H\varphi_n(X_{t_n})\Big] &= \mathbb{E}_P\Big[H g_P(X_{t_{n-1}})\Big] \\
&= \mathbb{E}_P\Big[H g_Q(X_{t_{n-1}})\Big] \\
&= \mathbb{E}_Q\Big[H g_Q(X_{t_{n-1}})\Big] \\
&= \mathbb{E}_Q\Big[H\varphi_n(X_{t_n})\Big],
\end{aligned}$$

where the third equality uses the induction hypothesis because the integrand depends only on $(X_{t_0}, \dots, X_{t_{n-1}})$. This proves equality of all finite-dimensional distributions. The Borel sigma-field on $C([u,T];\mathbb{R}^d)$ is generated by finite-dimensional coordinate maps, so $P = Q$. $\square$

Remark 2.4 (Why this is not a full solution)**.** *The theorem says that the probabilistic uniqueness problem and the Eulerian Fokker–Planck uniqueness problem are equivalent. It does not say when the Fokker–Planck equation is unique in terms of verifiable regularity, geometry, or algebraic structure of A and b. Therefore it is a correct replacement for the earlier overclaimed argument, but not a solution of the broad coefficient-characterization problem.*

3 Positive uniqueness results for discontinuous coefficients

We now prove coefficient-level results for two special classes. Throughout this section the coefficients are time-homogeneous. This restriction is used essentially in the elementary time-change proofs.

3.1 Isotropic covariance: arbitrary Borel speed

Theorem 3.1 (Isotropic Borel covariance)**.** *Let $a : \mathbb{R}^d \to \mathbb{R}$ be Borel and satisfy*

$$0 < \lambda \leq a(x) \leq \Lambda < \infty, \qquad x \in \mathbb{R}^d. \tag{6}$$

Then the martingale problem for

$$Lf(x) = \frac{1}{2} a(x) \Delta f(x), \qquad f \in C_c^\infty(\mathbb{R}^d), \tag{7}$$

is well posed for every initial law. Equivalently, the driftless SDE

$$\mathrm{d}X_t = \sqrt{a(X_t)}\, \mathrm{d}W_t$$

has uniqueness in law. The solution started from x is Brownian motion started from x under the time change

$$\Gamma_u = \int_0^u \frac{1}{a(B_r)}\, \mathrm{d}r, \qquad A_t = \Gamma_t^{-1}, \qquad X_t = B_{A_t}. \tag{8}$$

Proof. Let B be standard d-dimensional Brownian motion with $B_0 = x$. Because (6) holds, Γ in (8) is continuous, strictly increasing, and satisfies

$$\Lambda^{-1} u \leq \Gamma_u \leq \lambda^{-1} u.$$

Thus its inverse A is continuous and finite for all finite times. The time-changed process $X_t = B_{A_t}$ is continuous. By the optional time-change theorem applied to

$$f(B_u) - f(B_0) - \frac{1}{2} \int_0^u \Delta f(B_r)\, \mathrm{d}r, \qquad f \in C_c^\infty(\mathbb{R}^d),$$

and by the identity $A'_t = a(B_{A_t}) = a(X_t)$ for a.e. t, the process

$$f(X_t) - f(X_0) - \frac{1}{2} \int_0^t a(X_r) \Delta f(X_r)\, \mathrm{d}r$$

is a martingale. This proves existence.

For uniqueness, let X be any solution started from x. By applying the martingale problem to coordinate functions and products, using standard localization, $X - X_0$ is a continuous local martingale with quadratic covariation

$$\langle X^i - X_0^i, X^j - X_0^j \rangle_t = \delta_{ij} \int_0^t a(X_r)\, \mathrm{d}r.$$

Set

$$S_t = \int_0^t a(X_r)\,\mathrm{d}r, \qquad \tau_u = \inf\{t \geq 0 : S_t > u\}.$$

Again (6) implies that S is continuous, strictly increasing, and has a continuous inverse. The multidimensional Dambis–Dubins–Schwarz theorem, or equivalently Levy's characterization after time change, gives that

$$B_u := X_{\tau_u}$$

is standard d-dimensional Brownian motion started from x. The inverse-change formula for absolutely continuous increasing functions gives

$$\tau_u = \int_0^u \frac{1}{a(X_{\tau_r})}\,\mathrm{d}r = \int_0^u \frac{1}{a(B_r)}\,\mathrm{d}r.$$

Hence $\tau = \Gamma$ and $S = A = \Gamma^{-1}$. Therefore every solution has the representation $X_t = B_{A_t}$ with B Brownian, so its law is unique. Mixtures over an arbitrary initial law give well-posedness for all initial laws. □

Corollary 3.2 (Isotropic piecewise-constant coefficients)**.** *Let $\mathbb{R}^d$ be partitioned into arbitrary Borel sets E_k and let*

$$a(x) = \sum_k a_k \mathbf{1}_{E_k}(x), \qquad 0 < \lambda \leq a_k \leq \Lambda < \infty.$$

Then the driftless SDE $\mathrm{d}X_t = \sqrt{a(X_t)}\,\mathrm{d}W_t$ has uniqueness in law in every dimension. No smoothness or geometric regularity of the interfaces ∂E_k is required.

3.2 Coordinate-separable diagonal covariance

The next result is genuinely diagonal rather than scalar. Its proof uses the Dambis–Dubins–Schwarz theorem coordinate by coordinate, together with Knight's theorem that the Brownian motions obtained from pairwise strongly orthogonal continuous local martingales are independent.

Theorem 3.3 (Coordinate-separable diagonal covariance)**.** *Let $q_i : \mathbb{R} \to \mathbb{R}$, $i = 1, \ldots, d$, be Borel functions satisfying*

$$0 < \lambda \leq q_i(r) \leq \Lambda < \infty, \qquad r \in \mathbb{R}, \quad i = 1, \ldots, d. \tag{9}$$

Then the martingale problem for

$$Lf(x) = \frac{1}{2}\sum_{i=1}^d q_i(x_i)\partial_{ii} f(x), \qquad f \in C_c^\infty(\mathbb{R}^d), \tag{10}$$

is well posed for every initial law. Equivalently, the driftless diagonal SDE

$$\mathrm{d}X_t^i = \sqrt{q_i(X_t^i)}\,\mathrm{d}W_t^i, \qquad i = 1, \ldots, d, \tag{11}$$

has uniqueness in law.

Proof. First take deterministic initial condition $x \in \mathbb{R}^d$. Let $B^1, \ldots, B^d$ be independent standard one-dimensional Brownian motions starting from 0. For each i, define

$$\Gamma_i(u) = \int_0^u \frac{1}{q_i(x_i + B_r^i)}\,\mathrm{d}r, \qquad A_i(t) = \Gamma_i^{-1}(t), \qquad X_t^i = x_i + B_{A_i(t)}^i.$$

The bounds (9) make each Γ_i a continuous strictly increasing bijection of $[0, \infty)$ onto itself. Each X^i is a continuous local martingale with

$$\langle X^i \rangle_t = A_i(t), \qquad A_i'(t) = q_i(X_t^i) \quad \text{for a.e. } t.$$

Independence of the Brownian coordinates gives $\langle X^i, X^j \rangle \equiv 0$ for $i \neq j$. Ito's formula and localization then show that X solves the martingale problem for (10). This proves existence.

For uniqueness, let X be any solution of the martingale problem started from x. Localization of the coordinate and coordinate-product test functions gives continuous local martingales

$$M_t^i = X_t^i - x_i, \qquad i = 1, \dots, d,$$

with

$$\langle M^i, M^j \rangle_t = \delta_{ij} \int_0^t q_i(X_r^i) \, \mathrm{d}r. \tag{12}$$

In particular, the M^i are pairwise strongly orthogonal. Set

$$S_i(t) = \int_0^t q_i(X_r^i) \, \mathrm{d}r, \qquad \tau_i(u) = \inf\{t \geq 0 : S_i(t) > u\}.$$

By (9), each S_i is continuous, strictly increasing, and finite-valued with continuous inverse τ_i. Define

$$B_u^i = M_{\tau_i(u)}^i, \qquad u \geq 0.$$

The Dambis–Dubins–Schwarz theorem gives that each B^i is a standard one-dimensional Brownian motion. Knight's theorem applied to the pairwise strongly orthogonal martingales in (12) gives that $B^1, \dots, B^d$ are independent. Finally the inverse-change formula yields, for every i,

$$\tau_i(u) = \int_0^u \frac{1}{q_i(x_i + B_r^i)} \, \mathrm{d}r.$$

Thus $\tau_i = \Gamma_i$ and $S_i = A_i = \Gamma_i^{-1}$, so

$$X_t^i = x_i + B_{A_i(t)}^i, \qquad i = 1, \dots, d.$$

The joint law is therefore the law constructed in the existence part, and is unique.

For an arbitrary initial law ν, disintegrate any solution with respect to X_0. The conditional law given $X_0 = x$ must be the unique deterministic-start law just identified, for ν-a.e. x. Therefore the solution law is the unique mixture of deterministic-start laws against ν. $\square$

Corollary 3.4 (Coordinate-threshold and coordinate-strip coefficients)**.** *Let each q_i be an arbitrary bounded uniformly positive Borel function of one coordinate, for example a finite or countable step function*

$$q_i(r) = \sum_k q_{i,k} \mathbf{1}_{I_{i,k}}(r), \qquad 0 < \lambda \leq q_{i,k} \leq \Lambda < \infty,$$

for a Borel partition $\{I_{i,k}\}_k$ of $\mathbb{R}$. Then the diagonal SDE (11) has uniqueness in law in every dimension. In particular, multidimensional products of one-dimensional oscillating Brownian motions are weakly unique.

3.3 Adding bounded Borel drift by Girsanov

Proposition 3.5 (Bounded drift perturbation). *Let $A : \mathbb{R}^d \to \mathbb{S}^d_+$ be Borel, uniformly elliptic and bounded:*

$$\lambda I \le A(x) \le \Lambda I, \qquad x \in \mathbb{R}^d.$$

Assume that the driftless martingale problem

$$L^0 f(x) = \frac{1}{2} \operatorname{Tr}(A(x) D^2 f(x))$$

is well posed for every initial law. Then, for every bounded Borel drift $b : \mathbb{R}^d \to \mathbb{R}^d$, the martingale problem

$$L^b f(x) = b(x) \cdot \nabla f(x) + \frac{1}{2} \operatorname{Tr}(A(x) D^2 f(x)) \tag{13}$$

is well posed for every initial law. Consequently the corresponding Itô SDE has uniqueness in law for any Borel square root σ with $\sigma\sigma^\top = A$.

Proof. It suffices to treat deterministic initial condition x; arbitrary initial laws follow by disintegration. Let P^0_x be the unique driftless law and let $M_t = X_t - x$ be its canonical local martingale, with

$$\langle M \rangle_t = \int_0^t A(X_r) \, \mathrm{d}r.$$

Define

$$N_t = \int_0^t b(X_r)^\top A(X_r)^{-1} \, \mathrm{d}M_r.$$

Since b is bounded and $A \ge \lambda I$, the quadratic variation satisfies

$$\langle N \rangle_t = \int_0^t b(X_r)^\top A(X_r)^{-1} b(X_r) \, \mathrm{d}r \le \lambda^{-1} \|b\|_\infty^2 t.$$

Novikov's condition holds on every finite time interval. Hence the stochastic exponential $Z_t = \mathcal{E}(N)_t$ is a true martingale. Define P^b_x on $\mathcal{F}_T$ by $\mathrm{d}P^b_x = Z_T \, \mathrm{d}P^0_x$. Girsanov's theorem for continuous martingales gives that, under P^b_x,

$$X_t - x - \int_0^t b(X_r) \, \mathrm{d}r$$

is a continuous local martingale with quadratic variation $\int_0^t A(X_r) \, \mathrm{d}r$. Thus P^b_x solves the martingale problem for L^b.

Now let P be any solution of the L^b martingale problem from x. Put

$$\widetilde{M}_t = X_t - x - \int_0^t b(X_r) \, \mathrm{d}r$$

and define

$$\widetilde{N}_t = \int_0^t b(X_r)^\top A(X_r)^{-1} \, \mathrm{d}\widetilde{M}_r.$$

The same bound on quadratic variation implies that $\mathcal{E}(-\widetilde{N})_t$ is a true martingale. Under the probability Q with density $\mathcal{E}(-\widetilde{N})_T$ relative to P, Girsanov's theorem removes the drift, so Q solves the driftless martingale problem from x. Hence $Q = P^0_x$. Reversing the absolutely continuous change of measure gives $P = P^b_x$. Therefore the L^b martingale problem is unique. □

Corollary 3.6 (Two drifted discontinuous classes). *Let $b : \mathbb{R}^d \to \mathbb{R}^d$ be bounded and Borel. Weak uniqueness holds in every dimension for both classes below:*

(i) $\mathrm{d}X_t = b(X_t) \, \mathrm{d}t + \sqrt{a(X_t)} \, \mathrm{d}W_t$, *where $a : \mathbb{R}^d \to [\lambda, \Lambda]$ is arbitrary Borel;*

(ii) $\mathrm{d}X^i_t = b_i(X_t) \, \mathrm{d}t + \sqrt{q_i(X^i_t)} \, \mathrm{d}W^i_t$, *$i = 1, \dots, d$, where $q_i : \mathbb{R} \to [\lambda, \Lambda]$ are arbitrary Borel.*

Proof. Apply theorem 3.5 to theorem 3.1 and theorem 3.3, respectively. □

4 What remains unresolved for the ranking-game coefficients

The positive results above allow very rough discontinuities, including piecewise-constant coefficients with wild Borel interfaces in the isotropic case and arbitrary coordinate-strip discontinuities in the diagonal case. They do *not* cover the main diagonal structure suggested by finite ranking games, where the ith diagonal entry may depend on the rank or relative order of the full vector $x = (x_1, \dots, x_d)$. A schematic example is

$$A_{ii}(x) = \sigma_{\text{low}}^2 \mathbf{1}_{\{\text{rank}(x_i) \leq k\}} + \sigma_{\text{high}}^2 \mathbf{1}_{\{\text{rank}(x_i) > k\}}, \tag{14}$$

with discontinuities along collision hyperplanes $x_i = x_j$. Such coefficients are diagonal and piecewise constant, but they are not coordinate-separable: $A_{ii}(x)$ depends on other coordinates through the ordering of the system.

The Fokker–Planck criterion in theorem 2.3 says that weak uniqueness for (14) is equivalent to uniqueness of

$$\partial_t \mu_t = \frac{1}{2} \sum_{i=1}^{d} \partial_{ii} \big(A_{ii} \mu_t \big).$$

This is a correct formulation, but it leaves the actual interface analysis untouched. Across the hyperplanes $x_i = x_j$, distributional uniqueness would require controlling how probability flux is transmitted through lower-dimensional collision sets. The time-change proof for scalar coefficients fails because there is no common scalar clock, and the coordinate-separable proof fails because the ith clock depends on the other coordinates. A real solution of the ranking-game uniqueness problem would need a new argument for this coupled diagonal geometry, or a reduction to known well-posed reflected or rank-based particle systems with all label dynamics specified.

5 Conclusion

The earlier proposed note contained a correct idea but an incorrect conclusion. The correct idea is the martingale-problem/Fokker–Planck equivalence; the incorrect conclusion is that this equivalence solves the authors' coefficient-characterization problem. After repair, the result is best summarized as follows.

- The equivalence between weak uniqueness and Fokker–Planck determinacy is rigorous under standard superposition and conditioning hypotheses, but is a reformulation, not a coefficient-level answer.
- A genuine coefficient-level theorem is obtained for isotropic Borel uniformly elliptic covariance $A = aI$, with arbitrary bounded Borel drift.
- A second coefficient-level theorem is obtained for coordinate-separable diagonal covariance $A = \text{diag}(q_i(x_i))$, again with arbitrary bounded Borel drift.
- The diagonal rank/threshold systems motivating the paper remain outside the scope of these arguments, so this note gives partial progress only.

References

[1] S. Ankirchner, N. Kazi-Tani, J. Wendt, and C. Zhou. Large ranking games with diffusion control. *Mathematics of Operations Research*, 49(2):675–696, 2024.

[2] V. I. Bogachev, M. Röckner, and S. V. Shaposhnikov. On the Ambrosio–Figalli–Trevisan superposition principle for probability solutions to Fokker–Planck–Kolmogorov equations. *Journal of Dynamics and Differential Equations*, 33:715–739, 2021.

[3] S. N. Ethier and T. G. Kurtz. *Markov Processes: Characterization and Convergence.* Wiley, 1986.

[4] I. Karatzas and S. E. Shreve. *Brownian Motion and Stochastic Calculus.* Springer, second edition, 1991.

[5] N. Nadirashvili. Nonuniqueness in the martingale problem and the Dirichlet problem for uniformly elliptic operators. *Annali della Scuola Normale Superiore di Pisa, Classe di Scienze*, 24(3):537–550, 1997.

[6] D. Revuz and M. Yor. *Continuous Martingales and Brownian Motion.* Springer, third edition, 1999.

[7] D. W. Stroock and S. R. S. Varadhan. *Multidimensional Diffusion Processes.* Springer, 1979.

[8] D. Trevisan. Well-posedness of multidimensional diffusion processes with weakly differentiable coefficients. *Electronic Journal of Probability*, 21:1–41, 2016.